

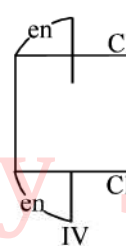
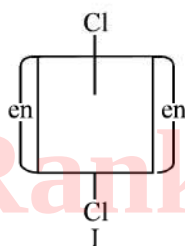
WORK SHEET - 02

COORDINATION COMPOUND

Single correct

- Q.1 How many geometrical isomers and stereoisomers are possible for $[\text{Pt}(\text{NO}_2)(\text{NH}_3)(\text{NH}_2\text{OH})(\text{Py})]^+$ and $[\text{Pt}(\text{Br})(\text{Cl})(\text{I})(\text{NO}_2)(\text{NH}_3)(\text{Py})]$ respectively
(A) 3 and 15 (B) 3 and 30 (C) 4 and 15 (D) 4 and 30
- Q.2 On treatment of 100 mL of 0.1 M solution of $\text{CoCl}_3 \cdot 6\text{H}_2\text{O}$ with excess AgNO_3 ; 0.6×10^{22} ions are precipitated. The complex is :
(A) $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$ (B) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$
(C) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ (D) $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
- Q.3 In which of the following pairs, hybridization, magnetic moment & geometry are same?
(A) $[\text{CrF}_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ (B) $[\text{IrF}_6]^{3-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$
(C) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{CN})_6]^{4-}$ (D) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{Ni}(\text{NH}_3)_4]^{2+}$
- Q.4 The crystal field stabilization energy of $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ is
(A) $7.2 \Delta_0$ (B) $0.4 \Delta_0$ (C) $2.4 \Delta_0$ (D) $3.6 \Delta_0$
- Q.5 In the complex acetyl bromidodicarbonylbis(triethylphosphine)iron(II), the number of Fe–C bond(s) is
(A) 3 (B) 4 (C) 6 (D) 8
- Q.6 What is oxidation state, magnetic moment and type of hybridisation of central metal cation in following complex : $[\text{Os}(\text{ONO})(\text{O})_2(\text{SCN})(\text{H}_2\text{O})]\text{OH}$
(A) +7, $\sqrt{3}$ B.M., d^2sp^3 hybridisation (B) +8, 0 B.M., sp^3d^2 hybridisation
(C) +8, 0 B.M., d^2sp^3 hybridisation (D) +9, 0 B.M., sp^3d^2 hybridisation
- Q.7 Which of the following statement is correct regarding the complex $[\text{CuL}_6]^{2+}$, one solution is blue and another is green.
(A) The blue solution will have higher value of Δ_0 .
(B) The green solution will have higher value of Δ_0 .
(C) Both have same value of Δ_0 .
(D) No prediction can be done regarding Δ_0 .
- Q.8 The correct order of magnetic moments is :
(A) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$ (B) $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
(C) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$ (D) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
- Q.9 Choose the correct code regarding, possible number of geometrical isomers exhibited by following complexes:
(I) $[\text{CrCl}_2(\text{NO}_2)_2(\text{NH}_3)_2]^-$ (II) $[\text{Co}(\text{NO}_2)_3(\text{NH}_3)_3]$
(III) $[\text{PtCl}(\text{NO}_2)(\text{NH}_3)(\text{py})]$ (IV) $[\text{PtBrCl}(\text{en})]$
- | | (I) | (II) | (III) | (IV) | | (I) | (II) | (III) | (IV) |
|-----|-----|------|-------|------|-----|-----|------|-------|------|
| (A) | 4 | 2 | 3 | 1 | (B) | 5 | 2 | 4 | 0 |
| (C) | 3 | 2 | 0 | 1 | (D) | 5 | 2 | 3 | 0 |

- Q.10 In the complex with formula $MCl_3 \cdot 4H_2O$, the coordination number of the metal M is six and there is no molecule of hydration in it. The volume of 0.1 M $AgNO_3$ solution needed to precipitate the free chloride ions in 200 mL of 0.01 M solution of the complex is :
 (A) 40 mL (B) 20 mL (C) 60 mL (D) 80 mL
- Q.11 50 ml of 0.2 M solution of a compound with empirical formula $CoCl_3 \cdot 4NH_3$ on treatment with excess of $AgNO_3(aq)$ yields 1.435 g of $AgCl$. Ammonia is not removed by treatment with concentrated H_2SO_4 . The formula of the compound is:
 (A) $CoCl_3(NH_3)_4$ (B) $[CoCl_2(NH_3)_4]Cl$
 (C) $[Co(NH_3)_4]Cl_3$ (D) $[CoCl_3(NH_3)]NH_3$
- Q.12 How many pairs of enantiomers are possible for following complex compound.
 $[M(AB)(CD)ef]^n\pm$
 (where AB, CD – Unsymmetrical bidentate ligand ; e,f – Monodentate ligands)
 (A) 20 (B) 5 (C) 10 (D) 8
- Q.13 The number of unpaired electrons calculated in $[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ are :
 (A) 4 and 4 (B) 0 and 2 (C) 2 and 4 (D) 0 and 4
- Q.14 Identify the geometrical isomers of the following:



- (A) I with III (B) II with IV (C) I with II and IV (D) none of these

- Q.15 Which of the following is not correctly matched?

Complex ion	CFSE
(A) $[Co(NH_3)_6]^{3+}$	24 Dq
(B) $[Cr(NH_3)_6]^{3+}$	12 Dq
(C) $[FeF_6]^{3-}$	4 Dq
(D) $[Fe(CN)_6]^{3-}$	20 Dq

- Q.16 Among the following pairs of complexes, in which case the Δ_0 value is higher for the first one?

(A) $[Co(NH_3)_6]^{3+}$ and $[Co(CN)_6]^{3-}$	(B) $[CoF_6]^{3-}$ and $[Co(NH_3)_6]^{3+}$
(C) $[Co(H_2O)_6]^{2+}$ and $[Co(H_2O)_6]^{3+}$	(D) $[Rh(H_2O)_6]^{3+}$ and $[Co(H_2O)_6]^{3+}$

- Q.17 The total number of possible isomers of the compound $[Cu^{II}(NH_3)_4][Pt^{II}Cl_4]$ are:

- (A) 3 (B) 5 (C) 4 (D) 6

- Q.18 In which of the following octahedral complexes of Co (at. no. 27), will the magnitude of Δ_0 be the highest?

- (A) $[Co(CN)_6]^{3-}$ (B) $[Co(C_2O_4)_3]^{3-}$ (C) $[Co(H_2O)_6]^{3+}$ (D) $[Co(NH_3)_6]^{3+}$

- Q.19 Assign the hybridization, shape and magnetic moment of $K_3[Cu(CN)_4]$:
 (A) sp^3 , tetrahedral, 1.73 B.M. (B) dsp^2 , square planar, 1.73 B.M.
 (C) sp^3 , tetrahedral, diamagnetic (D) dsp^2 , square planar, 2.44 B.M.
- Q.20 In the complexes $[Fe(H_2O)_6]^{3+}$, $[Fe(CN)_6]^{3+}$, $[Fe(C_2O_4)_3]^{3-}$ and $[FeCl_6]^{3-}$, more stability is shown by –
 (A) $[Fe(H_2O)_6]^{3+}$ (B) $[Fe(CN)_6]^{3+}$ (C) $[Fe(C_2O_4)_3]^{3-}$ (D) $[FeCl_6]^{3-}$
- Q.21 The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1 , L_2 , L_3 and L_4 absorb wavelengths in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is :
 (A) $L_1 < L_3 < L_2 < L_4$ (B) $L_3 < L_2 < L_4 < L_1$
 (C) $L_1 < L_2 < L_4 < L_3$ (D) $L_4 < L_3 < L_2 < L_1$
- Q.22 $[Fe^{II}Cl(CN)_4(O_2)]^{4-}$ is named as:
 (A) chloridotetracyanidodioxido ferrate (II) ion
 (B) chloridotetracyanidoperoxido ferrate (II) ion
 (C) chloridotetracyanidosuperoxido ferrate (II) ion
 (D) none is correct
- Q.23 Which of the following is an oxidizing agent?
 (A) $Mn(CO)_5$ (B) $Fe(CO)_5$ (C) $Mn_2(CO)_{10}$ (D) $Fe_2(CO)_9$
- Q.24 Consider the following complexes
 (I) $[Cr(CO)_x]$ (II) $[Cr(CO)_{(x-1)}PF_3]$
 If PF_3 is better π acceptor than CO, what will be the order of bond length of CO in complexes (I) and (II) :
 (A) $I > II$ (B) $II > I$ (C) $I = II$ (D) can not be compared
- Q.25 Which of the following statements is correct ?
 (A) $[CoF_6]^{3-}$ and $[Co(NH_3)_6]^{3+}$ both are paramagnetic complexes.
 (B) $[CoF_6]^{3-}$ and $[Co(NH_3)_6]^{3+}$ both are high spin complexes.
 (C) $[CoF_6]^{3-}$ is octahedral while $[Co(NH_3)_6]^{3+}$ has a pentagonal pyramid shape.
 (D) $[CoF_6]^{3-}$ is outerorbital complex while $[Co(NH_3)_6]^{3+}$ is inner orbital complex.
- Q.26 The equation which is balanced and represents the correct product(s) is :
 (A) $CdSO_4 + KCN \text{ (excess)} \longrightarrow K_3[Cd(CN)_4] + K_2SO_4$
 (B) $MgO + Al \longrightarrow Al_2O_3 + Mg$
 (C) $[CoCl(NH_3)_5]^+ + 5H^+ \longrightarrow Co^{2+} + 5NH_4^+ + Cl^-$
 (D) $[Mg(H_2O)_6]^{2+} + (EDTA)^{4-} \xrightarrow{\text{excess NaOH}} [Mg(EDTA)]^{2-} + 6H_2O$
- Q.27 An ion M^{2+} , forms the complexes $[M(H_2O)_6]^{2+}$, $[M(en)_3]^{2+}$ and $[MBr_6]^{4-}$, match the complex with the appropriate colour.
 (A) Green, blue and red (B) Blue, red and green
 (C) Green, red and blue (D) Red, blue and green
- Q.28 Which of the following ligands is called π -acceptors ?
 CO CN⁻ NO⁺
 (I) (II) (III)
 (A) I, II, III correct. (B) I, II only correct
 (C) II, III only correct (D) III only correct

Assertion Reason

- Q.29 **Statement-1** : $[\text{Fe}(\text{EDTA})]^-$ complex is octahedral in shape
Statement-2 : EDTA is a hexadentate ligand and undergoing sp^3d^2 hybridisation
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
- Q.30 **Statement-1** : In Zeise's salt C – C bond order regarding ethylene molecule is less than two.
Statement-2 : Ethylene accepts electron pair from filled d-orbital of Pt^{2+} into its vacant bonding M.O.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
- Q.31 **Statement-1**: Complexes containing three bidentate groups such as $[\text{Cr}(\text{ox})_3]^{3-}$ and $[\text{Co}(\text{en})_3]^{3+}$ do not show optical activity.
Statement-2: Octahedral complex, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ shows geometrical isomerism.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Comprehension

Paragraph for question nos. 32 to 34

$\text{Ni}^{II}(\text{NH}_3)_4(\text{NO}_3)_2 \cdot 2\text{H}_2\text{O}$ molecule may have two unpaired electrons or zero unpaired electron and measurement of magnetic moment helps us to predict the geometry.

- Q.32 If magnetic moment value is zero then the formula of the complex will be
(A) $[\text{Ni}(\text{NH}_3)_4](\text{NO}_3)_2 \cdot 2\text{H}_2\text{O}$ (B) $[\text{Ni}(\text{NH}_3)_2(\text{H}_2\text{O})_2](\text{NO}_3)_2 \cdot 2\text{NH}_3$
(C) $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2](\text{NO}_3)_2$ (D) $[\text{Ni}(\text{NO}_3)_2(\text{H}_2\text{O})_2]$
- Q.33 If the magnetic moment value is $2\sqrt{2}$ and conducts electricity then the formula of the complex is
(A) $[\text{Ni}(\text{NH}_3)_4](\text{NO}_3)_2 \cdot 2\text{H}_2\text{O}$ (B) $[\text{Ni}(\text{NH}_3)_2(\text{H}_2\text{O})_2](\text{NO}_3)_2 \cdot 2\text{NH}_3$
(C) $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2](\text{NO}_3)_2$ (D) $[\text{Ni}(\text{NH}_3)_4(\text{NO}_3)_2] \cdot 2\text{H}_2\text{O}$
- Q.34 The higher and lower value of magnetic moment of the given complex corresponds to the following geometries respectively.
(A) Octahedron and Tetrahedron (B) Octahedron and square planar
(C) Square planar and Octahedron (D) Octahedron and Octahedron



Paragraph for question nos. 35 & 36

The crystal field theory (C.F.T.) is now much more widely accepted than the valence bond theory. It assumes that the attraction between the central metal and the ligands in a complex is purely electrostatic. According to C.F.T. ligands are treated as a point charge and crystal field splitting energy (CFSE) increases the thermodynamic stability of the complexes. Value of CFSE depends upon nature of ligand and a spectrochemical series has been made experimentally. For tetrahedral complexes, Δ is about $4/9$ times to Δ_0 (CFSE for octahedral complexes). This energy lies in visible region and i.e. why electronic transitions are responsible for colour.

- Q.35 Crystal field stabilization energy for $[\text{CoF}_6]^{3-}$ is
(A) $0.6 + P$ (B) $-0.4 + P$ (C) $1.2 + 2P$ (D) $2.4 + 4P$
[P is pairing energy]
- Q.36 $\text{Ti}^{3+}(\text{aq})$ is violet while $\text{Ti}^{4+}(\text{aq})$ is colourless because -
(A) There is no crystal field effect in Ti^{4+}
(B) There energy difference between t_{2g} and e_g of Ti^{4+} is quite high and does not fall in the visible region.
(C) Ti^{4+} has d^0 configuration.
(D) Ti^{4+} is very small in comparison to Ti^{3+} and hence does not absorb any radiation.

More than one may be correct

- Q.37 Choose **incorrect** statement(s) regarding following complex ion.
 $[\text{Fe}(\text{ox})_3]^{3-}$ and $[\text{Fe}(\text{NO}_2)_6]^{3-}$
(A) $[\text{Fe}(\text{ox})_3]^{3-}$ complex ion is more stable than $[\text{Fe}(\text{NO}_2)_6]^{3-}$.
(B) Both complex ions are optically inactive.
(C) Both follow Sidwick's rule of E.A.N.
(D) Both are paramagnetic.
- Q.38 When O_2 is adsorbed on a metallic surface, electron transfer occurs from the metal to O_2 . The **TRUE** statement(s) regarding this adsorption is(are)
(A) O_2 is physisorbed (B) heat is released
(C) occupancy of π_{2p}^* of O_2 is increased (D) bond length of O_2 is increased.
- Q.39 Select **incorrect** statement(s) for $[\text{Cu}(\text{CN})_4]^{3-}$, $[\text{Cd}(\text{CN})_4]^{2-}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ complex ion.
(A) Both $[\text{Cd}(\text{CN})_4]^{2-}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ have square planar geometry
(B) $[\text{Cu}(\text{CN})_4]^{3-}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ have equal no. of unpaired electron
(C) $[\text{Cu}(\text{CN})_4]^{3-}$ and $[\text{Cd}(\text{CN})_4]^{2-}$ can be separated from the mixture on passing H_2S gas.
(D) All the three complexes have magnetic moment equal to zero.
- Q.40 Which of the following statements is **not true** about the complex ion $[\text{CrCl}(\text{NO}_2)(\text{en})_2]^+$ (en = ethylene diamine)
(A) It has two geometrical isomers-cis and trans
(B) cis and trans forms are not diastereomers to each other
(C) Only the cis isomer displays optical activity
(D) It has three optically active isomers : d, l and meso forms
- Q.41 Which of the following compound(s) show(s) optical isomerism.
(A) $[\text{Pt}(\text{bn})_2]^{2+}$ (B) $[\text{CrCl}_2(\text{en})_2]^+$ (C) $[\text{Co}(\text{en})_3]$ (D) $[\text{Zn}(\text{gly})_2]$

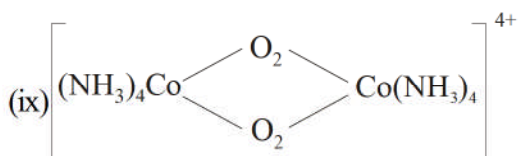
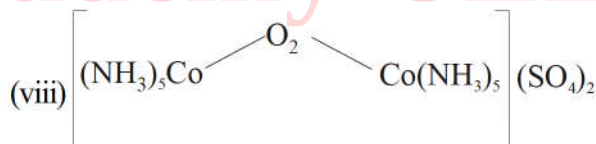
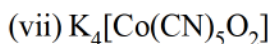
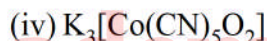
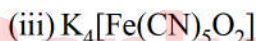
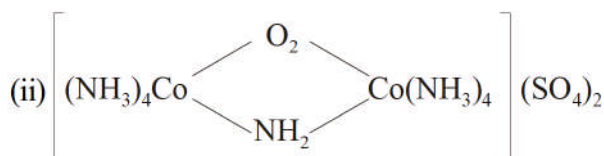
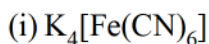


Match the column

Q.42	Column I	Column II
(A)	Only four stereoisomer	(P) $[M(AB)_3]^{n\pm}$
(B)	Four optically active isomer	(Q) $[M(AA)a_2b_2]^{n\pm}$
(C)	Double the number of geometrical isomer compared to any other complex given in column II.	(R) $[M a_2b_2 cd]^{n\pm}$
		(S) $[Ma_3bcd]^{n\pm}$
Q.43	Column I	Column II
(A)	$[Ma_2b_2c_2]^{n\pm}$	(P) four optically active isomers
(B)	$[Ma_3b_2c]^{n\pm}$	(Q) total 15 space isomers
(C)	$[Ma_2bcde]^{n\pm}$	(R) three geometrical isomers
(D)	$[M(AB)_3]^{n\pm}$	(S) except one, other geometrical isomers are optically inactive.

Subjective

Q.44 Find the number of paramagnetic species each central metal ion having $t_{2g}^6 e_g^0$ configuration.



Q.45 If all six positions of octahedral geometry are marked by different colours first, then any one bidentate unsymmetrical ligand having no chiral centre can be introduced by _____ number of ways.

Q.46 Write the sum of Geometrical isomers of $[Pt(H_2N-CH(CH_3)-COO^-)_2]$ complex and stereo isomers of $[Pt(gly)_3]^+$ complex.

Q.47 The number of optically active isomer for $[Pt(NH_3)_2(F)(Cl)(Br)(I)]^0$ is _____.

Q.48 Find the number of t_{2g} and e_g electrons in $[NiF_6]^{2-}$.
[If ans. is 4 & 2 then represent as 42]

ANSWER KEY

Q.1	B	Q.2	D	Q.3	B	Q.4	C	Q.5	A
Q.6	C	Q.7	A	Q.8	A	Q.9	D	Q.10	B
Q.11	B	Q.12	C	Q.13	D	Q.14	C	Q.15	C
Q.16	D	Q.17	C	Q.18	A	Q.19	C	Q.20	B
Q.21	A	Q.22	C	Q.23	A	Q.24	A	Q.25	D
Q.26	C	Q.27	B	Q.28	A	Q.29	C	Q.30	C
Q.31	D	Q.32	A	Q.33	C	Q.34	B	Q.35	B
Q.36	C	Q.37	ABC	Q.38	BCD	Q.39	ABD	Q.40	BD
Q.41	ABCD	Q.42	(A) P, Q (B) P, R (C) R, S	Q.43	(A) S, (B) R, (C) Q, (D) P				
Q.44	0004	Q.45	0024	Q.46	0008	Q.47	0012	Q.48	0060

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